Mathematics & Statistics | Lancaster University

Introducing the Trend Locally Stationary Wavelet Process

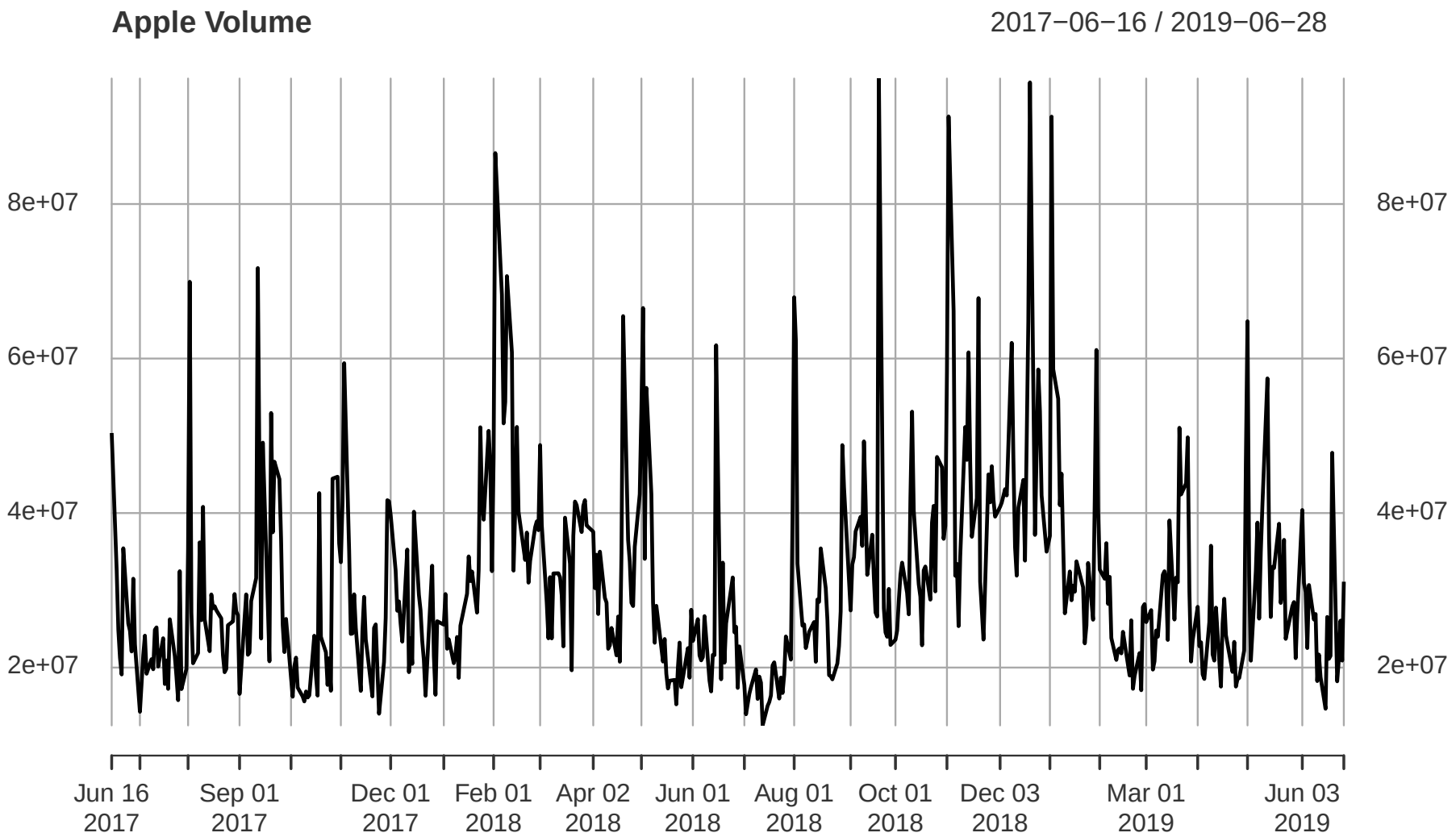
Rebecca Killick(r.killick@lancs.ac.uk)
Seminar Open Uni 2020

Outline

- Motivation
- Locally Stationary Wavelet Processes
- Introducing the T-LSW process
- Fitting a T-LSW process
- Simulations
- Data Example

Motivation

Mathematics & Statistics | Lancaster University



Current Approach

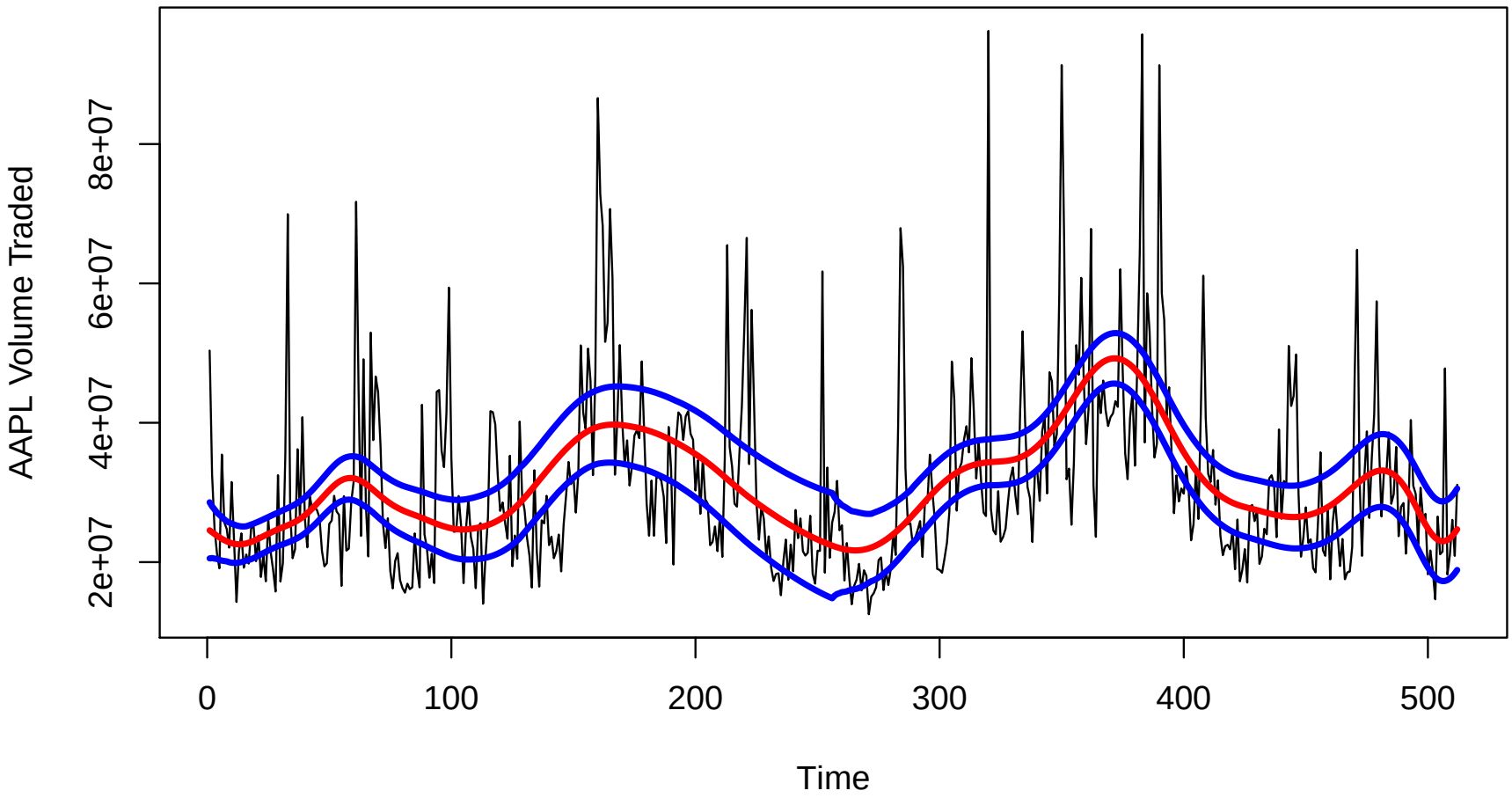
- Typically a focus is on (nonstationary) first OR second order structure
- If first order then the second order is assumed constant and/or known (von Sachs, MacGibbon, 2000)
- If second order then the mean is assumed to be 0 or removed (Nason, 2013)
  - Often via first differencing (Killick et al., 2013)

Exception: Tunyavetchakit, 2010, unpublished PhD Thesis

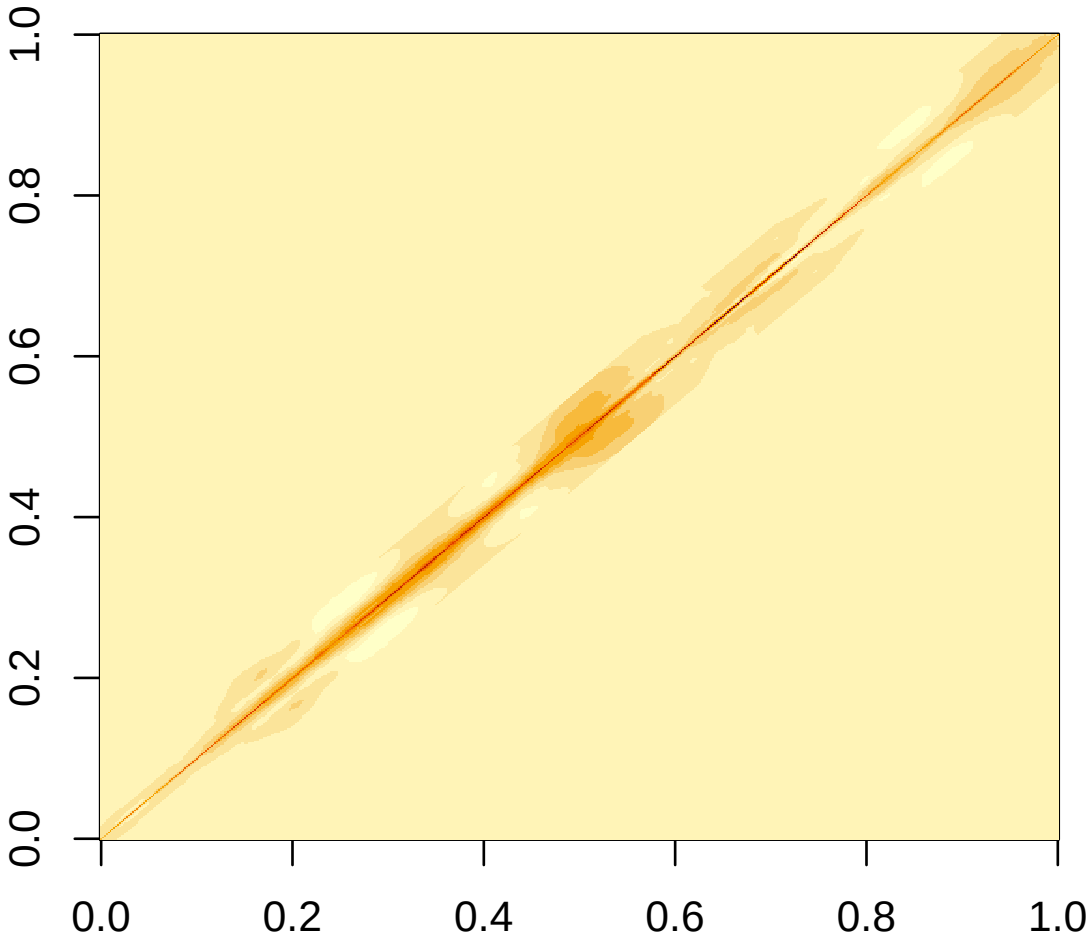
. . . needs more investigation – Dahlhaus Review (2012)

In Essence . . .

. . . the method i'm proposing gives:

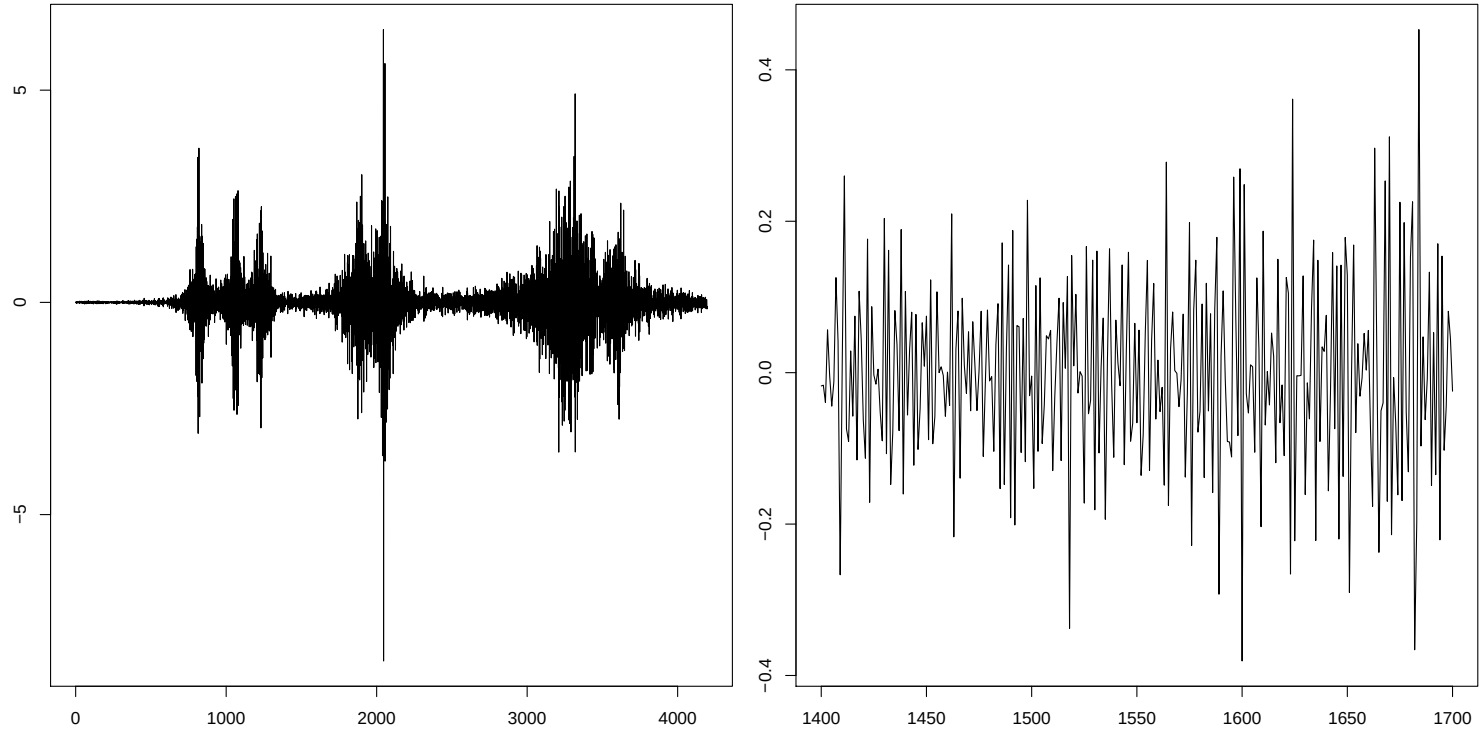


In Essence . . .



Locally stationary models

If you take a small enough region, it will appear stationary as the structure varies slowly over time.



Application areas include;

- medicine, finance
- environmental processes, e.g. wind speeds.

Locally Stationary Wavelet

Locally Stationary Wavelet Model (Nason et al. 2000)

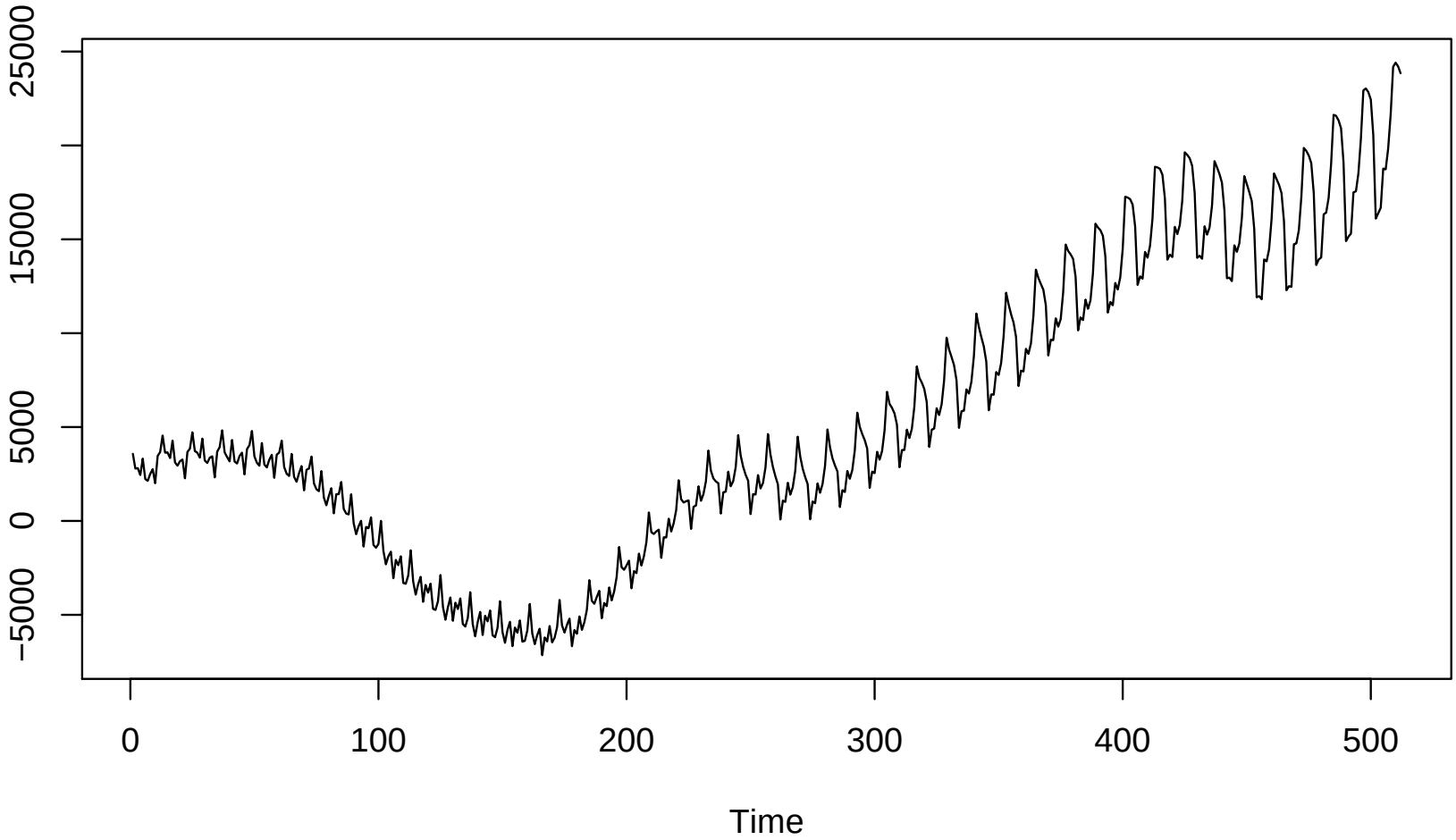
$$X_{t,n} = \sum_j \sum_k w_j \left(\frac{k}{n}\right) \psi_{j,k-t} \xi_{jk}.$$

Relevant assumptions:

- $\mathbb{E}(\xi_{jk}) = 0$, $\text{cov}(\xi_{jk}, \xi_{lm}) = \delta_{jl}\delta_{km}$
- Hence $\mathbb{E}(X_t) = 0$, i.e. remove any mean or trend before analysis
- The $w_j^2(\cdot)$ are assumed close to a Lipschitz continuous function $W_j(z)$.
- LSW processes encapsulate other models and represent processes whose variance and autocorrelation function vary over time.

Example - Data

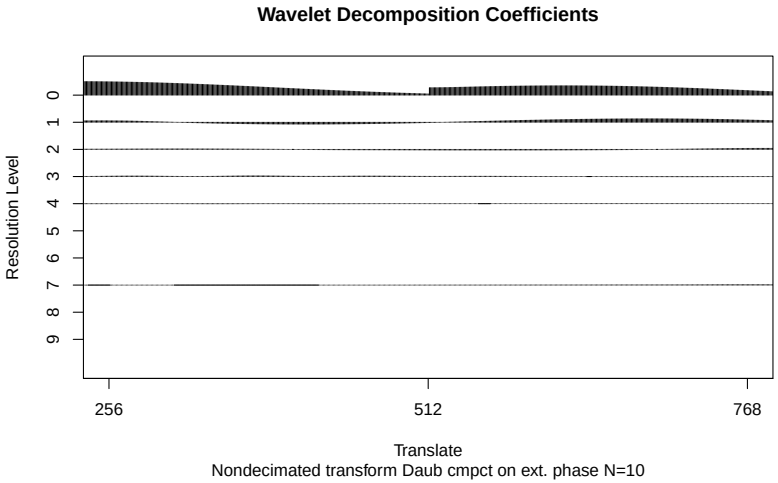
SARIMA(1,1,1)(1,1,1)[12]



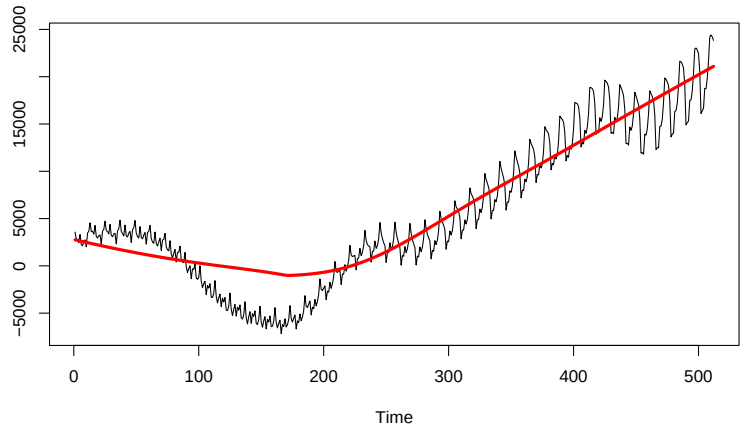
Example - Analysis

Mathematics & Statistics | Lancaster University

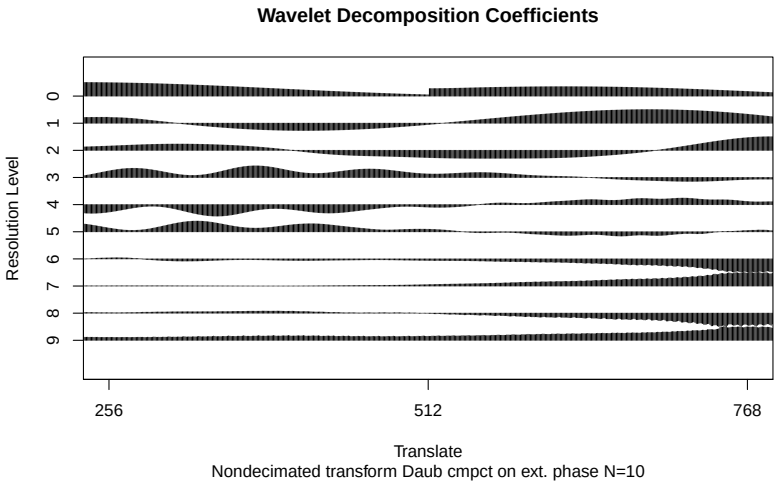
Periodogram



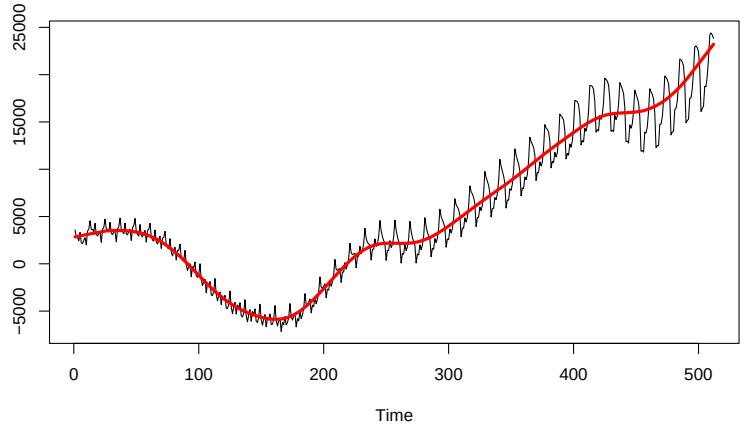
Loess default fit



Periodogram scaling by level



Loess small smoothing span



T-LSW model

A nonstationary time series can be modelled as a trend locally stationary wavelet (T-LSW) process $\{X_{t,T}\}$, $t = 0, ..., T-1$, and $T = 2^J$, as follows:

$$X_{t,T} = \mu(t/T) + \sum_{j=-\infty}^{-1} \sum_{k \in \mathbb{Z}} w_j(k/T)\psi_{j,k-t}\xi_{j,k},$$

where

- μ is a low order polynomial mean function,
- $w_j(k/T)$ are amplitudes, the smoothness of which controls the degree of nonstationarity,
- $\{\psi_{j,k-t}\}_{j,k}$ is a set of discrete non-decimated wavelets,
- $\{\xi_{j,k}\}$ is a sequence of orthonormal random variables.

Remarks

- This extends the rigorous theory of the LSW model of Nason et al. (2000) to include a trend.
- The model uses rescaled time $z = k/T \in (0, 1)$.
- This means we use in-fill asymptotic theory.
- Amplitudes W_j are assumed to be Lipschitz continuous as before.
- Low order polynomial assumption can be relaxed.

The wavelet spectrum

The key second order quantity of interest is the wavelet spectrum, defined as:

$$S_j(z) = |W_j(z)|^2, \text{ for } z = k/T \in (0,1).$$

- It measures the contribution to variance at a particular rescaled time z and scale j.
- This is a decomposition of the autocovariance across timescales.

Spectrum estimation

We estimate the spectrum $S_j(k/T)$ using the wavelet periodogram $I_k^j = |d_{j,k}|^2$, where

$$d_{j,k} := \langle X_{t,T}, \psi_{j,k-t} \rangle = \sum_t X_{t,T} \psi_{j,k-t}$$

are the empirical wavelet coefficients.

Assuming a lower order polynomial trend with respect to the wavelet,

$$\hat{S}_j(k/T) = \sum_{l=-J}^{-1} A_{jl}^{-1} \tilde{I}_k^l.$$

Smoothing and correcting yields an asymptotically unbiased, consistent estimate.

Local autocovariance

The local autocovariance (LACV) is given by

$$c(z,\tau) = \sum_j S_j(z)\Psi_j(\tau),$$

where $\Psi_j(\tau)$ is the autocorrelation wavelet at scale j, lag τ.

We estimate this using

$$\hat{c}(z,\tau) = \sum_{j=-J_0}^{-1} \hat{S}_j(z)\Psi_j(\tau),$$

where $J_0 = \alpha \log_2(T)$ for $\alpha \in (0,1)$ is the coarsest scale used.

Estimator is consistent.

Trend estimation procedure

- Use linear wavelet thresholding, since the wavelet coefficients of the trend will be zero:

$$\hat{\mu}(t/T) = \sum_j \sum_k \theta_{j,k} \psi_{j,k-t},$$

  where $\theta_{j,k} = I(j < J_0) d_{j,k}$, J_0 is the coarsest scale of the wavelet transform.

- Keep boundary coefficients as well as they contain trend.
- Estimator is consistent provided time series is short-memory.
- Works well in practice for non-polynomial trends too.

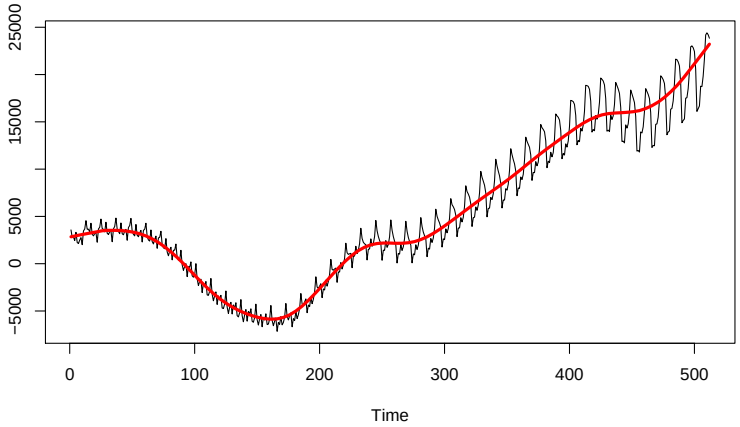
Less smooth trends

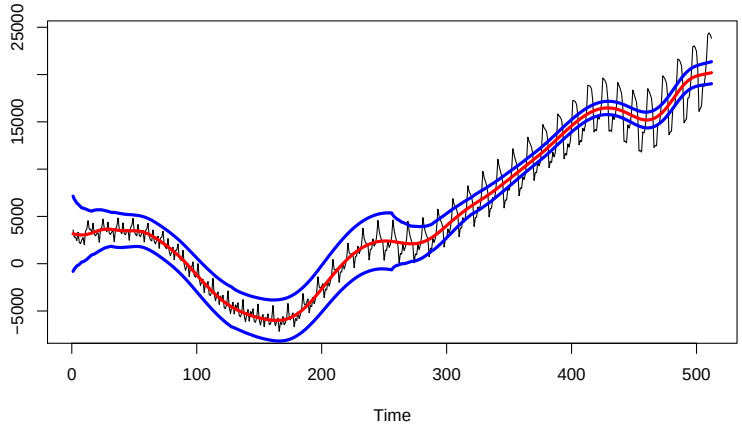
- Low-order polynomial trend may not be realistic.
- For a high-order polynomial we can use a higher order wavelet and correct the spectral estimate with a different matrix.
- For a general smooth trend (Hölder continuous), we use a tapered estimator:

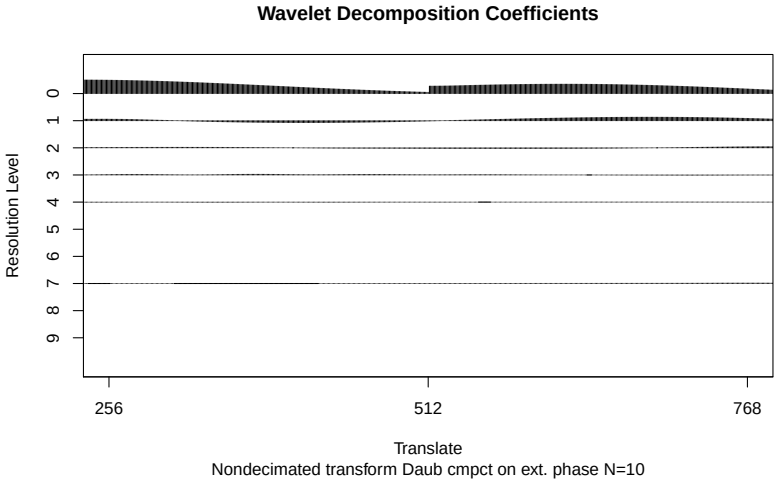
$$\hat{S}_j(k/T) = \sum_{l=-J_0}^{-1} A_{jl}^{-1} \tilde{I}_k^l, \; J_0 < J.$$

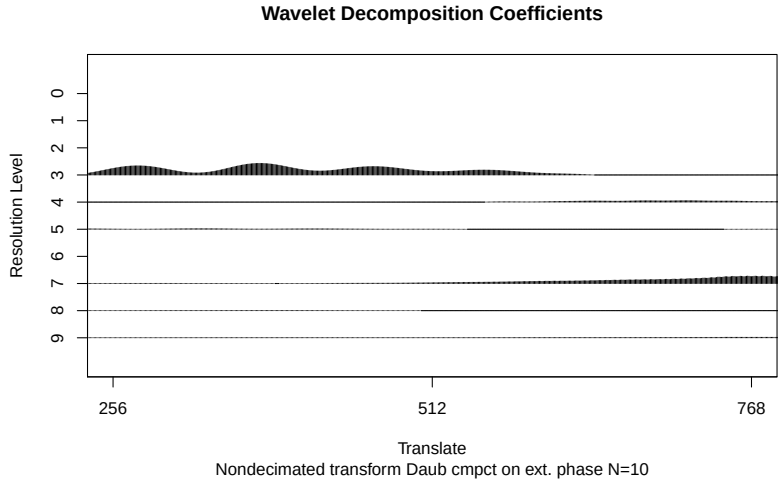
- Estimate LACV in same way as before.
- Estimators are consistent using running mean smoothing.

Trend Estimation

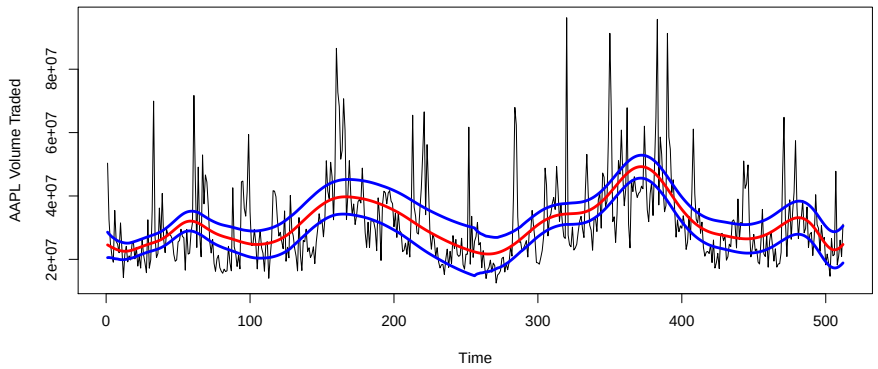


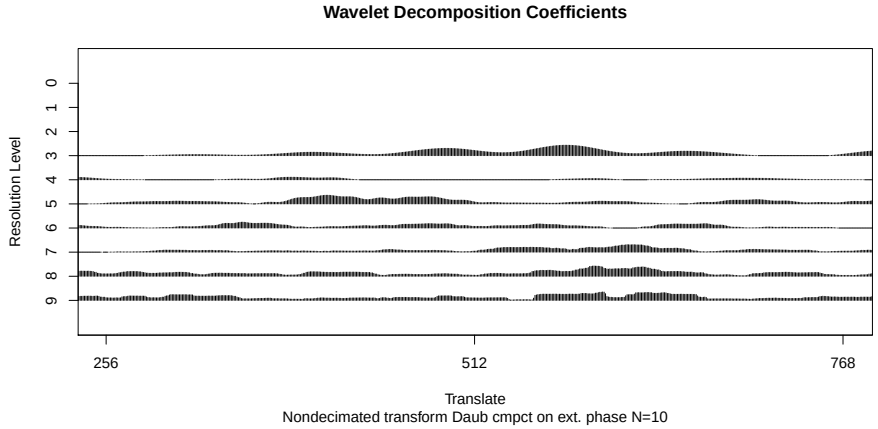


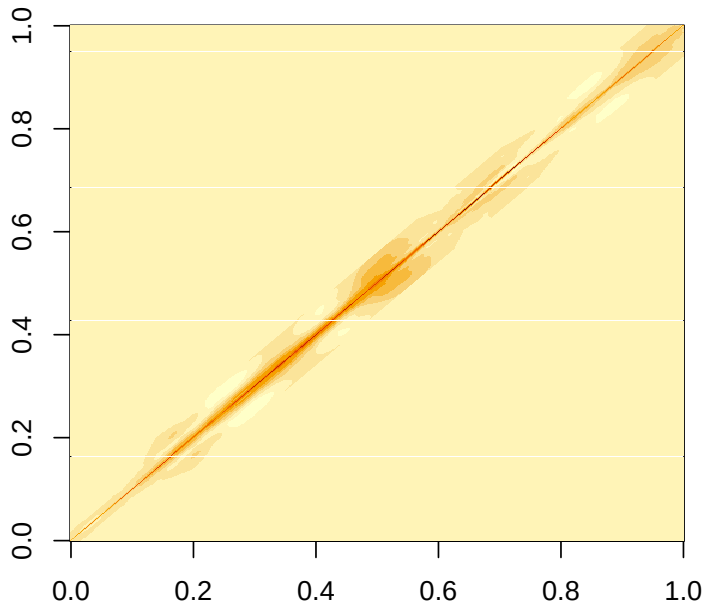




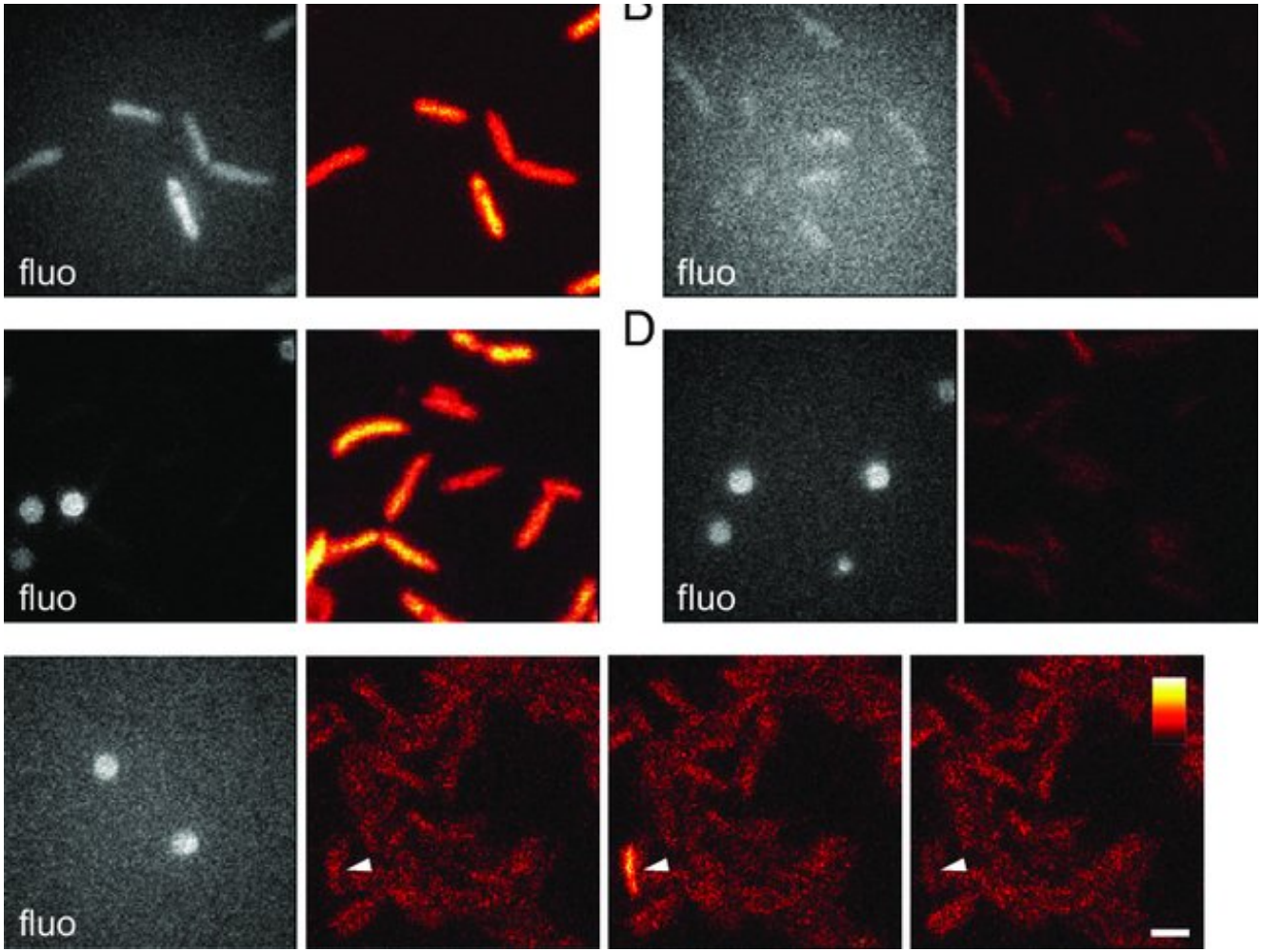
AAPL Volume Traded



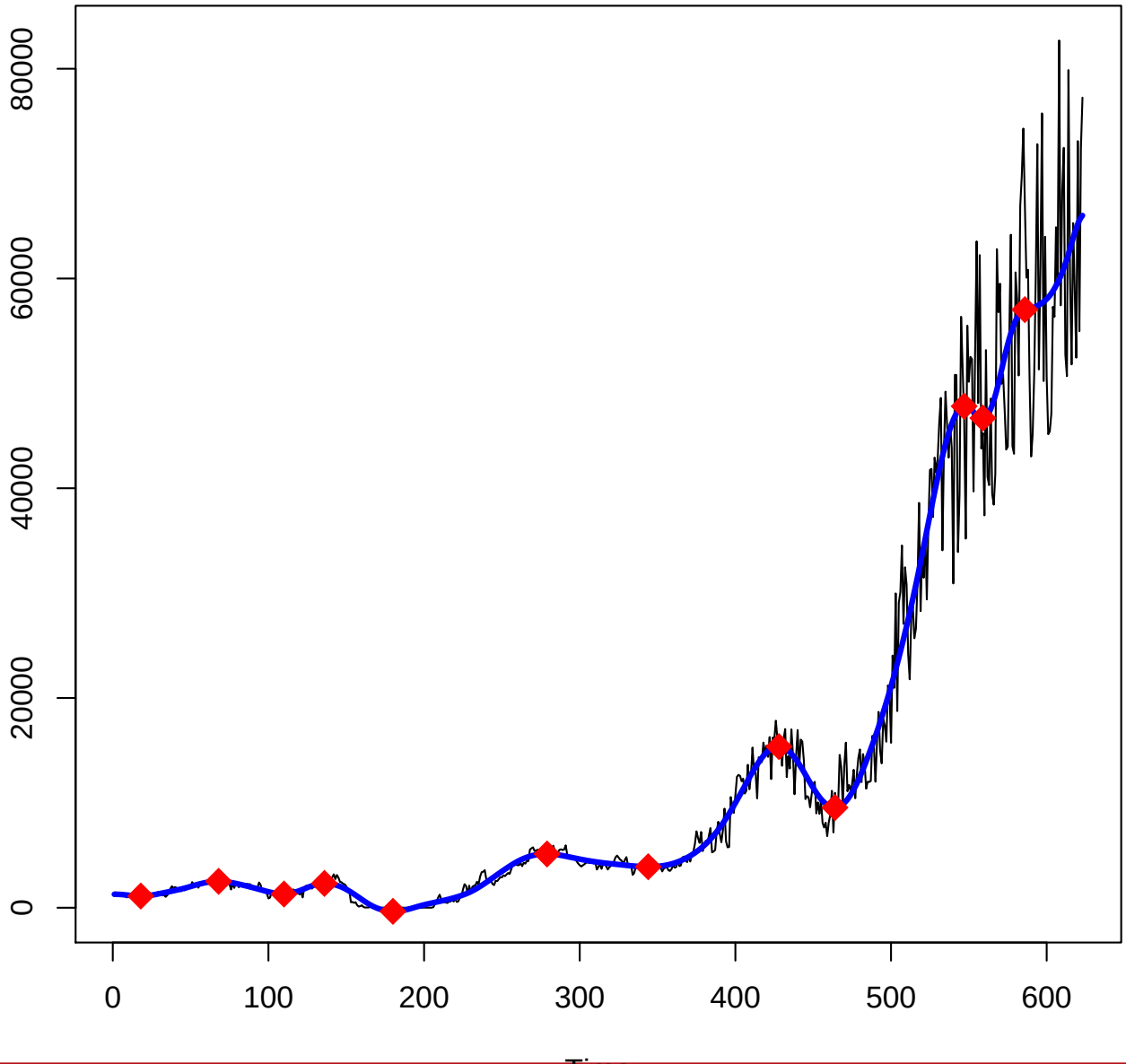




Bioluminescence



Bioluminescence



Summary

- Introduced the T-LSW process (+ rigorous theory in paper)
- Demonstrated how to fit it
- Simulations indicated sensible performance (more detail in paper)
- Useful in practice too

Future work:

- Extension to total variation bounds on the spectra
- Derive theoretical results for applications of the model to forecasting etc.